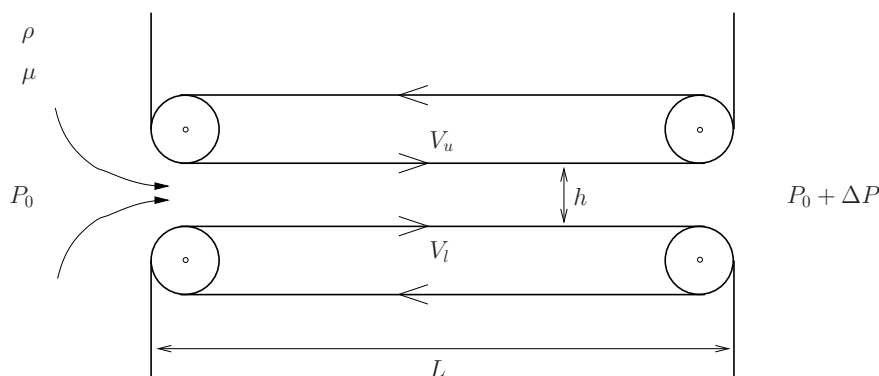


PROBLEM 3 (50 minutes)

The system sketched in the figure is designed to steadily pump fluid of constant density ρ and constant viscosity μ from a reservoir, where the modified pressure has a constant value P_0 , to another one at pressure $P_0 + \Delta P$ ($\Delta P > 0$). To that end, a pair of parallel belts of length L , separated by a distance $h \ll L$, are driven at constant velocities V_l and V_u , dragging the liquid towards the large pressure reservoir.

In the forthcoming analysis, the liquid flow can be assumed steady and two-dimensional (spanwise length much larger than h). The most important parameters of the system are the liquid volume flux **per unit spanwise length**, q , the power needed to drive the system **per unit spanwise length**, $\dot{W}$, and the pump efficiency, $\eta = q\Delta P/\dot{W}$.



1. (3 points). Use dimensional analysis to simplify the dependence of q , $\dot{W}$ and η , as functions of an appropriately chosen set of parameters governing the operation of the pump.
2. (2 points). Simplify the resulting dimensionless functions when the inertia of the fluid can be neglected compared to the viscous forces. Note that, in this limit, the fluid density plays a negligible role. In the remainder of the problem, assume that these conditions are satisfied.

A pump with $h = 1$ cm and $L = 20$ cm is calibrated using glycerol, of density $\rho_g = 1260$ kg/m³ and viscosity $\mu_g = 1.41$ kg/(m s), using an overpressure $\Delta P_g = 1$ kPa. During the calibration process, the lower belt is kept at rest, $V_{l,g} = 0$, while the upper belt is driven at several velocities $V_{u,g}$, as indicated in the table below, where the corresponding values of q_g , $\dot{W}_g$ and η_g are also shown.

$V_{u,g}$ (cm/s)	q_g (cm ² /s)	$\dot{W}_g$ (Watt/m)	η_g
10	2	0.8	0.26
15	4.5	1.4	0.33
20	7	2.1	0.33
25	9.5	3	0.32
30	12	4	0.3
35	14.5	5.2	0.28

The **same pump** is used to deliver a silicone oil, of density $\rho_{oil} = 1100$ kg m⁻³ and viscosity $\mu_{oil} = 5$ kg/(m s), with an overpressure $\Delta P_{oil} = 2$ kPa.

3. (2 points). Determine the velocities $V_{l,oil}$ and $V_{u,oil}$ for which the results obtained in the calibration tests can be used to predict the pumping behavior with the silicone oil.
4. (2 points). Calculate the corresponding values of q_{oil} , $\dot{W}_{oil}$ and η_{oil} .
5. (1 point). Discuss the hypothesis of negligible fluid inertia.

$$\textcircled{1} \quad q = f_1(s, \bar{\mu}, \Delta P, \bar{V}_u, \bar{V}_e, L, h)$$

$$\dot{W} = f_2(s, \mu, \Delta P, V_u, V_e, L, h)$$

$$\eta = f_3(s, \mu, \Delta P, V_u, V_e, L, h)$$



(μ, V_u, h) ARE DIMENSIONALLY INDEPENDENT. APPLYING Π -THEOREM:

$$\boxed{\begin{aligned} \frac{q}{V_u h} &= \pi_q \left(\frac{s V_u h}{\mu}, \frac{h \Delta P}{\mu V_u}, \frac{V_e}{V_u}, \frac{L}{h} \right) \\ \frac{\dot{W}}{\mu V_u^2} &= \pi_w \left(\downarrow, \downarrow, \downarrow, \downarrow \right) \\ \eta &= \eta \left(\downarrow, \downarrow, \downarrow, \downarrow \right) \end{aligned}}$$

$$\textcircled{2} \quad \text{LIMIT } \frac{s V_u h}{\mu} \rightarrow 0 \Rightarrow \begin{cases} \frac{q}{V_u h} = \pi_q \left(\frac{h \Delta P}{\mu V_u}, \frac{V_e}{V_u}, \frac{L}{h} \right) \\ \frac{\dot{W}}{\mu V_u^2} = \pi_w \left(\frac{h \Delta P}{\mu V_u}, \frac{V_e}{V_u}, \frac{L}{h} \right) \\ \eta = \eta \left(\frac{h \Delta P}{\mu V_u}, \frac{V_e}{V_u}, \frac{L}{h} \right) \end{cases}$$

$\textcircled{3}$ PHYSICAL SIMILARITY: $\frac{L}{h}$ TAKES THE SAME VALUE (SAME PUMP)

$$\text{THEN } \frac{V_{e, \text{oil}}}{V_{u, \text{oil}}} = \frac{V_{e, g}}{V_{u, g}} = 0 \Rightarrow \boxed{V_{e, \text{oil}} = 0}$$

$$\frac{h \Delta P_{\text{oil}}}{\mu_{\text{oil}} V_{u, \text{oil}}} = \frac{h \Delta P_g}{\mu_g V_{u, g}} \Rightarrow \boxed{\frac{V_{u, \text{oil}}}{V_{u, g}} = \frac{\mu_g}{\mu_{\text{oil}}} \frac{\Delta P_{\text{oil}}}{\Delta P_g} = \frac{1.41}{.5} \frac{10}{5} \approx 0.564}$$

$$\textcircled{4} \quad \frac{q_{\text{oil}}}{V_{u, \text{oil}} h} = \frac{q_g}{V_{u, g} h} \Rightarrow \boxed{\frac{q_{\text{oil}}}{q_g} = \frac{V_{u, \text{oil}}}{V_{u, g}} \approx 0.564} ; \quad \boxed{\frac{\dot{W}_{\text{oil}}}{\dot{W}_g} = \frac{\mu_{\text{oil}}}{\mu_g} \left(\frac{V_{u, \text{oil}}}{V_{u, g}} \right)^2 \approx 1.128}$$

$V_{u, g}$ (m/s)	$V_{u, \text{oil}}$	q_{oil}	$\dot{W}_{\text{oil}}$	η_{oil}
10	5.64	1.128	0.902	0.26
15	8.46	2.538	1.579	0.33
20	11.28	3.948	2.369	0.33
25	14.1	5.358	3.384	0.32
30	16.92	6.768	4.512	0.3
35	19.74	8.178	5.866	0.28

$$\textcircled{5} \quad 0.045 \leq \frac{s_2 V_{u, g} h}{\mu_g} \leq 0.156$$

$$\left(0.0062 \leq \frac{s_{\text{oil}} V_{u, \text{oil}} h}{\mu_{\text{oil}}} \leq 0.022 \right)$$

↓
GOOD APPROXIMATION